

Operator Discovery Report: det_restart_controller_two_cluster_bottleneck

- Operator type: restart_controller.
- Host scaffold: nearest_neighbor_2opt.
- Core idea: Deterministic restart scheduling that triggers perturbation-based restarts under stagnation, then grows restart count conservatively to improve transfer on bottleneck/clustered EUC instances without changing the core loc
- Novelty classification: adaptive_restart_controller.
- Rediscovery signature: restart_controller:2:7:1.
- Surviving candidate: False.
- Problem class where it helps: two_cluster_bottleneck_tsp.

Pseudocode

- Start from a conservative restart budget.
- Increase restart count only after the configured stagnation trigger.
- Keep perturbation restarts bounded by a deterministic maximum.

Complexity

- Non-empty lines: 12.
- Complexity score: 0.44.
- Fingerprint: 211c74474fbe83b2.

Why It Should Help

- The operator changes restart pressure under stagnation, which is a compact way to regulate exploration without rewriting search internals.

Failure Cases

- Does not transplant cleanly into sparse_three_opt.
- Runtime inflation is high at 0.616565.

Transplant Results

- nearest_neighbor_2opt gap delta 0.0, runtime delta 139.778629, distance-eval delta 5471.428571.
- cheapest_insertion_2opt gap delta -0.002814, runtime delta 676.948914, distance-eval delta 4510.285714.
- random_restart_2opt gap delta 0.0, runtime delta -157.765486, distance-eval delta -10183.428571.
- sparse_three_opt gap delta 0.036996, runtime delta -71.189429, distance-eval delta -219.857143.
- clustered_local_search gap delta 0.001517, runtime delta -19.241857, distance-eval delta -7277.428571.

Instance-Family Results

- heldout_tsplib operator gap 0.235058 vs baseline 0.235058.
- clustered_tsp operator gap 0.110431 vs baseline 0.110853.

- elongated_corridor_tsp operator gap 0.003204 vs baseline 0.003204.
- grid_like_tsp operator gap 0.106696 vs baseline 0.106696.
- nearest_neighbor_trap_tsp operator gap 0.0 vs baseline 0.0.
- two_cluster_bottleneck_tsp operator gap 0.084632 vs baseline 0.109057.
- uniform_euclidean operator gap 0.0 vs baseline 0.0.

Pareto Results

- Gap delta: -0.003304.
- Runtime inflation: 0.616565.
- Distance-evaluation inflation: 0.763692.
- Same-gap-faster flag: False.
- Better-gap-same-runtime flag: False.

Ablation Results

- disabled mean gap 0.121317 and delta vs full operator 0.003304.
- fixed_restart_budget mean gap 0.118013 and delta vs full operator 0.0.
- wider_restart_budget mean gap 0.117688 and delta vs full operator -0.000325.